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CUET UG

Biology

Reproductive Health

Official Syllabus Unit VI · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET UG Biology 2026 / 2027 Baseline)	Exam Skills & Question Types
Reproductive Health: Strategies	Need for reproductive health and prevention of sexually transmitted diseases (STDs); birth control — need and methods, contraception and medical termination of pregnancy (MTP).	WHO holistic definition; RCH milestones; demographic growth factors; amniocentesis diagnostic scope vs statutory legal ban.
Contraceptive Modalities	Birth control — need and methods, contraception; natural, barrier, IUDs, oral pills, injectables, implants, surgical sterilization.	Classifications; copper vs hormone-releasing IUDs; **Saheli** non-steroidal weekly pill (CDRI); 72-hour emergency window; vasectomy vs tubectomy.
MTP & Amniocentesis	Medical Termination of Pregnancy (MTP); amniocentesis; statutory regulations.	MTP Act 1971 / 2017 amendments (12-week vs 20-week vs 24-week rules); karyotype identification of aneuploidies; female foeticide ban.
Sexually Transmitted Infections	Prevention of sexually transmitted diseases (STDs / RTIs); causative pathogens, clinical signs, and preventive measures.	Bacterial (**Neisseria**, **Treponema**) vs Viral; high-yield incurable triad (**Hepatitis B**, **Genital Herpes**, **HIV**); PID complications.
Infertility & ART Protocols	Infertility and assisted reproductive technologies — IVF, ZIFT, GIFT, ICSI, IUI (Artificial Insemination).	Clinical diagnostic thresholds; blastomere cutoffs (≤ 8 blastomeres $\rightarrow$ ZIFT vs > 8 $\rightarrow$ IUT); GIFT in vivo route; ICSI micromanipulation.

EXAM PATTERN

Official CUET UG Biology Pattern (NTA 2026/2027 Baseline): The Biology domain paper comprises **50 compulsory multiple-choice questions** to be answered within **60 minutes** (+5 for correct answers, −1 for incorrect responses). Unit VI (Reproductive Health) is an ultra-high-yield applied biology chapter testing precise diagnostic thresholds, pharmacological contraceptive mechanisms, statutory acts, and embryo transfer rules.

Module 1: Reproductive Health: Concepts, Problems & Public Strategies

According to the **World Health Organisation (WHO)**, reproductive health signifies **total well-being across all aspects of reproduction**, encompassing physical, emotional, behavioral, and social dimensions. A reproductively healthy society consists of individuals possessing structurally and

functionally normal reproductive organs and balanced, mature emotional-behavioral interactions in all sex-related matters.

India's Public Health Leadership & National Initiatives

- **Pioneering Global Leadership (1951):** India was the **first country in the world** to initiate nation-wide action plans and programs at a governmental level to achieve total reproductive health as a national social goal.
  - **Family Planning Programme:** Initiated in **1951** and periodically reassessed over subsequent decades.
  - **RCH Programmes:** Currently operationalized under the broader, modernized framework of **Reproductive and Child Health Care (RCH)** programmes.
- **Primary Objectives of RCH Programmes:**
  - Creating widespread public awareness regarding reproduction-related aspects via mass media (audio-visual, print, governmental and non-governmental agencies).
  - Providing infrastructure, clinical facilities, and professional support for building a reproductively healthy society.
  - Introducing comprehensive **sex education in schools** to dispel myths, misconceptions, and provide reliable, scientifically accurate information to adolescents.
  - Educating youth about reproductive anatomy, pubertal physiological changes, hygienic menstrual practices, and safe, responsible sexual behaviour.
  - Informing fertile couples and marriageable age cohorts about available birth control options, prenatal care of pregnant mothers, post-natal care of mother and infant, importance of **breastfeeding**, and equal opportunities for the male and female child.
  - Educating populations on the hazards of uncontrolled population explosion, sex-abuse, and sex-related crimes.

Amniocentesis: Diagnostic Scope vs Statutory Ban

Amniocentesis is a prenatal diagnostic procedure executed between the 14th and 18th weeks of gestation under ultrasonic guidance:

- **Biochemical & Cytological Basis:** A hypodermic needle is carefully inserted through the maternal abdominal and uterine walls into the amniotic sac to withdraw a small aliquot of **amniotic fluid**. The fluid contains shed **foetal skin, epithelial, and fibroblastic cells**.
- **Legitimate Diagnostic Scope:**
  - **Chromosomal Aberrations (Karyotyping):** Culturing foetal cells allows definitive diagnostic detection of genetic and chromosomal disorders such as **Down syndrome (47, +21)**, **Turner syndrome (45, XO)**, **Klinefelter syndrome (47, XXY)**, and Philadelphia chromosome translocation.
  - **Inborn Errors of Metabolism:** Detecting metabolic disorders such as **phenylketonuria (PKU)**, alkaptonuria, and cystic fibrosis.
  - **Hemoglobinopathies:** Screening for **sickle-cell anemia** and thalassemia.
  - **Foetal Maturity:** Quantifying lecithin-to-sphingomyelin ratio to assess pulmonary surfactant maturity.
- **Severe Misuse & Statutory Ban:**
  - Because foetal karyotyping reveals sex chromosomes (XX vs XY), the technique was extensively abused to determine the sex of the foetus, frequently followed by female foeticide.
  - **Pre-Conception and Pre-Natal Diagnostic Techniques (PCPNDT) Act, 1994:** India enacted a stringent statutory ban prohibiting the use of prenatal diagnostic

techniques for sex determination, imposing severe penal consequences to safeguard the female child and restore balanced national sex ratios.

Module 2: Population Explosion, Demographics & Birth Control

At the dawn of the 20th century (1900), the world population stood at approximately 2 billion (2000 million). By the year 2000, it exploded to 6 billion, and surpassed 7.2 billion by 2011. A similar exponential surge characterized India: from roughly 350 million at independence (1947), the population crossed 1 billion by May 2000 and exceeded 1.2 billion by 2011.

Drivers of the Demographic Transition

- **Primary Catalysts of Population Growth:**
  - **Rapid Decline in Death Rate:** Substantial improvements in medical infrastructure, sanitation, mass vaccination, and epidemic containment.
  - **Decline in Maternal Mortality Rate (MMR):** Safer obstetrics and increased institutional hospital deliveries.
  - **Decline in Infant Mortality Rate (IMR):** Eradication of neonatal tetanus, widespread oral rehydration therapy, and pediatric immunization.
  - **Surge in Reproductive Age Cohort:** A massive proportion of the population entering peak reproductive maturity.
- **Governmental Counter-Measures:**
  - Promoting small families through media slogans (**"Hum Do Hamare Do"** — We Two, Our Two).
  - Many urban, educated couples adopting a **"One Child Norm"**.
  - Raising the statutory legal marriageable age: **18 years for females** and **21 years for males**.
  - Incentivizing couples with small families through fiscal and employment benefits.

Essential Criteria of an Ideal Contraceptive

An ideal contraceptive device must fulfill the following clinical benchmarks:

1. **User-friendly** and easily accessible to all socio-economic strata.
2. **Highly effective** with minimal or no adverse physiological side effects.
3. **Reversible**, allowing prompt restoration of fertility upon discontinuation.
4. **Non-interfering** with sexual drive, desire, or the normal sexual act of the user.

Module 3: Contraceptive Modalities & Mechanisms of Action

Birth control methods prevent unwanted conceptions through five primary physiological axes: preventing gamete contact, suppressing ovulation, altering the uterine environment, blocking sperm transport, or surgical severance.

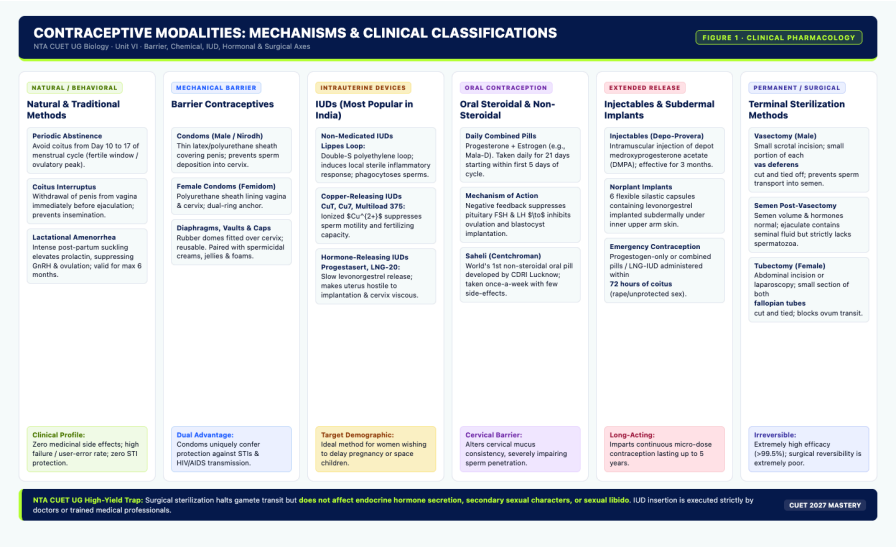


Figure 1: Systematic taxonomy, pharmacological modes of action, clinical indications, and failure profiles across natural, barrier, IUD, hormonal, and permanent contraceptive modalities.

1. Natural / Traditional Methods

Natural methods avoid the encounter of sperm and ovum without utilizing external devices or chemicals:

- **Periodic Abstinence (Rhythm Method):**
  - Couples avoid or abstain from coitus from **Day 10 to Day 17 of the menstrual cycle**.
  - **Physiological Rationale:** Ovulation occurs around Day 14 (± 2 days). The released ovum survives 24 hours, while viable sperms persist up to 72 hours in the female genital tract. Hence, Days 10–17 represent the **fertile window** of maximal conception probability.
- **Coitus Interruptus (Withdrawal):**
  - The male partner withdraws his penis from the vagina just prior to ejaculation, avoiding intra-vaginal insemination.
  - **Drawback:** High failure rate due to pre-ejaculatory emission containing viable spermatozoa.
- **Lactational Amenorrhea:**
  - Based on the physiological fact that ovulation and menstrual cycles do not occur during the period of intense, exclusive infant breastfeeding post-parturition.
  - **Endocrine Mechanism:** Suckling stimulates maternal hypothalamic prolactin-releasing factors, maintaining high **Prolactin**. Hyperprolactinemia suppresses pulsatile **GnRH** release, inhibiting pituitary LH and FSH surges required for follicular maturation and ovulation.
  - **Efficacy Window:** Highly effective **only up to a maximum period of 6 months post-partum**, provided feeding is frequent and unsupplemented. Zero medicinal side effects, but chance of failure rises thereafter.

2. Barrier Contraceptives

Barriers mechanically prevent ovum and sperm from physically meeting:

- **Condoms:** Sheaths fabricated from thin latex rubber or polyurethane:
  - **Male Condom (e.g., Nirodh):** Rolled over the erect penis immediately prior to coitus; traps ejaculate.
  - **Female Condom (Femidom):** Flexible pouch with two rings inserted into the vagina, covering cervix and vulva.

- **Dual Clinical Advantage:** Disposable, self-insertable (ensures user privacy), and **uniquely provides proven protection against Sexually Transmitted Infections (STIs) and HIV/AIDS**.

- **Diaphragms, Cervical Caps & Vaults:**
  - Rubber barriers inserted into the female reproductive tract to cover the cervix during coitus.
  - **Reusable:** Must be washed and stored; block sperm entry into the uterus.
  - **Chemical Synergism:** Routinely paired with **spermicidal creams, jellies, and foams** containing nonoxynol-9 to enhance contraceptive efficacy.

3. Intrauterine Devices (IUDs)

IUDs are devices inserted into the uterine cavity by certified medical practitioners or trained nurses through the cervix. They represent the **most widely accepted and popular method of contraception in India** for females seeking to delay pregnancy or space children:

- **A. Non-Medicated IUDs:**
  - **Lippes Loop:** Double-S-shaped polyethylene device impregnated with barium sulfate (for radiopacity).
  - **Mechanism:** Acts as a foreign body, inducing sterile local endometrial inflammation; massively attracts macrophages that **phagocytose spermatozoa** entering the uterine cavity.
- **B. Copper-Releasing IUDs:**
  - **Devices: CuT**, **Cu7**, and **Multiload 375**.
  - **Mechanism:** Continuously elute ionized copper (Cu^{2+}) into uterine secretions. Copper ions:
    1. Suppress sperm motility and flagellar kinematics.
    2. Inhibit sperm acrosomal enzyme activity, destroying fertilizing capacity.
    3. Increase phagocytosis of sperms within the uterus.
- **C. Hormone-Releasing IUDs:**
  - **Devices: Progestasert** and **LNG-20** (releases $20\mu g$ levonorgestrel/day).
  - **Mechanism:** Maintain a high localized intrauterine progestin gradient that:
    1. Induces profound endometrial atrophy, rendering the uterus **unsuitable for blastocyst implantation**.
    2. Thickens cervical mucus, forming a tenacious plug **hostile to sperm penetration**.
    3. Exerts mild local suppression of tubal motility.

4. Oral Contraceptive Pills

- **Combined Daily Oral Pills (e.g., Mala-D, Mala-N):**
  - Composed of synthetic progestogen combined with small amounts of estrogen (ethinyl estradiol).
  - **Administration Protocol:** Taken daily for **21 consecutive days**, commencing preferably within the first 5 days of the menstrual cycle. Followed by a 7-day pill-free interval (during which withdrawal bleeding occurs), after which the cycle is repeated.
  - **Triple Mode of Action:**
    1. **Ovulation Inhibition:** High circulating progestogen-estrogen exerts negative feedback on the hypothalamus and anterior pituitary, abolishing the mid-cycle LH surge.

2. **Implantation Block:** Alters the histology of the endometrium, preventing decidualization.
3. **Cervical Mucus Alteration:** Renders cervical secretions thick, tacky, and impenetrable to sperms.

- **Saheli (Centchroman / Ormeloxifene):**
  - The world's first **non-steroidal oral contraceptive pill**, developed by scientists at the **Central Drug Research Institute (CDRI) in Lucknow, India**.
  - **Pharmacological Mechanism:** Acts as a selective estrogen receptor modulator (SERM), competitively antagonizing endometrial estrogen receptors and desynchronizing blastocyst-endometrial dialogue to prevent implantation.
  - **Dosing Schedule:** Taken twice a week for the first 3 months, followed by **once a week** thereafter.
  - **Clinical Benefits:** Highly effective, non-hormonal, devoid of steroidal side effects (no weight gain, nausea, or fluid retention), with exceptional contraceptive value.

5. Injectables & Subdermal Implants

- **Injectables (e.g., Depo-Provera):** Intramuscular injections of depot medroxyprogesterone acetate (DMPA, 150 mg) administered every 3 months; provides continuous anovulatory contraception.
- **Subdermal Implants (e.g., Norplant):** A set of 6 flexible silastic capsules containing levonorgestrel implanted under the skin of the inner upper arm; slowly releases progestogen over a **5-year duration**.
- **Emergency Contraception:** Administration of progestogens alone, progestogen-estrogen combinations, or insertion of an IUD within **72 hours of unprotected coitus**. Extremely effective in preventing conception resulting from rape, contraceptive failure (e.g., condom rupture), or unprotected intercourse.

6. Surgical / Terminal Sterilization Methods

Sterilization methods represent permanent terminal procedures suitable for couples who have achieved their desired family size. They block gamete transport, preventing conception with extreme efficacy ($> 99.5\%$):

- **Vasectomy (Male Sterilization):**
  - A small incision is made in the scrotum, and a small segment of both **vasa deferentia** is excised and ligated (cut and tied).
  - **Semen Characteristics Post-Vasectomy:** The ejaculate contains normal volume of secretions from seminal vesicles, prostate, and bulbourethral glands, but is **completely devoid of spermatozoa (azoospermia)**.
  - **Clearance Period:** Sperms remain stored in the ampullae of the vas deferens and epididymis for several weeks; sterility is confirmed only after approximately 20 subsequent ejaculations or a negative semen analysis.
- **Tubectomy (Female Sterilization):**
  - A small portion of both **fallopian tubes** is excised and ligated through a minor incision in the abdomen or via laparoscopy through the vagina.
  - Blocks ovum transport; ovulated secondary oocytes cannot reach the ampulla and degenerate.
- **CUET High-Yield Trap: Sterilization procedures do NOT inhibit spermatogenesis, oogenesis, or gonadal steroid hormone production (testosterone/estrogen).** Secondary

sexual traits, erectile function, and libido remain completely unaffected. Surgical reversibility, however, is extremely poor.

WORKED EXAMPLE — COMPARATIVE CONTRACEPTIVE PHARMACOLOGY

Clinical Matching Problem: Match the contraceptive device in Column I with its precise primary mechanism of action in Column II:

Contraceptive Device	Primary Physiological Mechanism
1. Lippes Loop	**A.** Elutes Cu^{2+} ions to suppress sperm motility and fertilizing ability
2. LNG-20	**B.** Promotes macrophage-mediated phagocytosis of sperms in uterine cavity
3. Multiload 375	**C.** Non-steroidal SERM taken once weekly to inhibit implantation
4. Saheli	**D.** Releases levonorgestrel, rendering cervix hostile and endometrium atrophic

Correct Pairing: $1-B$, $2-D$, $3-A$, $4-C$.

Module 4: Medical Termination of Pregnancy (MTP) & Statutory Laws

Medical Termination of Pregnancy (MTP), or **induced abortion**, is the intentional or voluntary termination of pregnancy before the foetus achieves independent viability. Globally, nearly **45 to 50 million MTPs** are performed annually, accounting for approximately **one-fifth (20%) of the total number of conceived pregnancies**. While MTP plays a significant demographic role in decreasing population growth, its fundamental medical purpose is preventing maternal mortality resulting from high-risk or forced pregnancies.

The Statutory Evolution of the MTP Act

- **The MTP Act, 1971:** India was among the early nations to legalize induced abortion under specified clinical and humanitarian conditions. The primary intent was curbing maternal mortality caused by septicaemia and uterine perforation during illegal abortions performed by untrained quacks.
- **The MTP (Amendment) Act, 2021 (Latest NCERT & CUET Baseline):**
  - **Gestational Limit up to 20 Weeks:** Termination can be performed on the medical opinion of **one Registered Medical Practitioner (RMP)** formed in good faith.
  - **Gestational Limit from 20 to 24 Weeks:** Permitted for defined vulnerable categories of women, requiring the concurring opinions of **two Registered Medical Practitioners**. Vulnerable categories include:
    1. Survivors of rape, sexual assault, or victims of incest.
    2. Minors (below 18 years of age).
    3. Women whose marital status changed during ongoing pregnancy (widowhood or divorce).
    4. Women with physical disabilities or mental illness.
    5. Foetuses diagnosed with substantial physical or mental abnormalities.
  - **Beyond 24 Weeks:** Permitted exclusively for cases of **substantial foetal abnormalities** diagnosed by a statutory **State-level Medical Board** (comprising a gynaecologist, paediatrician, radiologist/sonologist, and other notified experts).

- **Universal Inclusivity:** The 2021 amendment explicitly replaced the word "married woman or her husband" with **"any woman or her partner"**, thereby granting legal abortion rights to unmarried females in cases of contraceptive failure.

- **Medical vs Surgical Abortion:**
  - **Medical Abortion (Early 1st Trimester — up to 7 to 9 Weeks):** Administration of the antiprogestin **Mifepristone (RU-486)** to block progesterone receptors and induce decidual necrosis, followed 48 hours later by the prostaglandin E_1 analog **Misoprostol** to stimulate powerful uterine contractions and expel the conceptus.
  - **Surgical Abortion (Later Stages):** Vacuum aspiration, dilation and curettage (D&C), or dilation and evacuation (D&E) performed under strict sterile operating theatre conditions.
- **CUET High-Yield Trap: Abortion is considered medically safest during the first trimester (up to 12 weeks of pregnancy).** Terminations executed in the second trimester are substantially more hazardous due to deep placental vascularization and maternal tissue fragility.

Module 5: Sexually Transmitted Infections (STIs / RTIs)

Infections or diseases transmitted primarily through sexual intercourse are collectively designated as **Sexually Transmitted Infections (STIs)**, **Venereal Diseases (VD)**, or **Reproductive Tract Infections (RTIs)**.

The Etiological Spectrum of STIs

Disease	Causative Pathogen	Class	Clinical Manifestations & Curability
Gonorrhoea	**Neisseria gonorrhoeae**	Bacterium	Gram-negative diplococcus; purulent urethral/cervical discharge, dysuria; **Curable** with cephalosporin antibiotics.
Syphilis	**Treponema pallidum**	Spirochete	Painless hard chancre (Primary) → skin rash (Secondary) → gummas & neurosyphilis (Tertiary); **Curable** with Penicillin.
Chlamydiasis	**Chlamydia trachomatis**	Bacterium	Obligate intracellular pathogen; mucopurulent cervicitis; major silent cause of PID; **Curable** with azithromycin.
Trichomoniasis	**Trichomonas vaginalis**	Protozoan	Flagellated parasite; foul-smelling, greenish-yellow frothy discharge, intense vulval pruritus; **Curable** with metronidazole.
Genital Warts	**Human Papillomavirus (HPV)** (Types 6, 11)	DNA Virus	Cauliflower-like fleshy condylomata on vulva/penis; oncogenic Types 16 & 18 induce cervical carcinoma; **Incurable** (treatable).
Genital Herpes	**Herpes Simplex Virus (HSV-2)**	DNA Virus	Painful fluid-filled vesicular ulcers on genitalia; latent in lumbosacral

Disease	Causative Pathogen	Class	Clinical Manifestations & Curability
			sensory ganglia; **Completely Incurable**.
Hepatitis B	**Hepatitis B Virus (HBV)**	DNA Virus	Acute jaundice, chronic cirrhosis, hepatocellular carcinoma; blood & sexual transmission; **Completely Incurable**.
AIDS	**Human Immunodeficiency Virus**	Retrovirus	Destruction of CD4+ T-helper lymphocytes, profound immune collapse, opportunistic infections; **Completely Incurable**.

Clinical Vulnerability, Complications & Public Prevention

- **CUET High-Yield Trap — The Incurable Viral Triad:** Except for **Hepatitis B**, **Genital Herpes**, and **HIV/AIDS**, all other STIs are **completely curable if detected early and treated adequately**.
- **Dual Modes of Transmission:** Both **Hepatitis B and HIV** are transmitted sexually as well as non-sexually via: (1) sharing of contaminated hypodermic needles/syringes among intravenous drug users; (2) transfusion of contaminated blood/blood products; and (3) transplacental mother-to-child transmission.
- **Peak Vulnerability Demographic:** Young adults in the **age group of 15 to 24 years** show the highest epidemiological incidence of STIs due to exploratory behaviour and inadequate protective knowledge.
- **Long-Term Complications of Untreated STIs:** Early localized symptoms (itching, watery or purulent fluid discharge, mild pelvic discomfort, burning micturition) are frequently ignored due to social stigma. Untreated infections ascend into the upper genital tract, triggering:
  1. **Pelvic Inflammatory Disease (PID)** — Chronic infection of ovaries, fallopian tubes, and pelvic peritoneum.
  2. **Ectopic Pregnancies** — Tubal scarring and strictures trapping blastocysts in the oviduct.
  3. **Infertility** — Irreversible bilateral tubal occlusion or azoospermia.
  4. **Adverse Pregnancy Outcomes** — Abortions, premature deliveries, and stillbirths.
- **National Prevention Guidelines:** (1) Avoid sex with unknown or multiple partners; (2) Consistently and correctly utilize condoms during every coital act; (3) In case of doubt, consult qualified medical specialists immediately for complete diagnostic screening and curative therapy.

Module 6: Infertility & Assisted Reproductive Technologies (ART)

Infertility is defined clinically as the **inability of a couple to achieve clinical pregnancy despite 1 to 2 years of regular, cohabitation with unprotected sexual intercourse**. Infertility stems from congenital, hormonal, physical, immunological, drug-induced, or psychological etiologies. While cultural biases in India historically ostracized the female partner, comprehensive reproductive audits show that **in over 50% of infertile couples, the physiological cause resides in the male partner**.

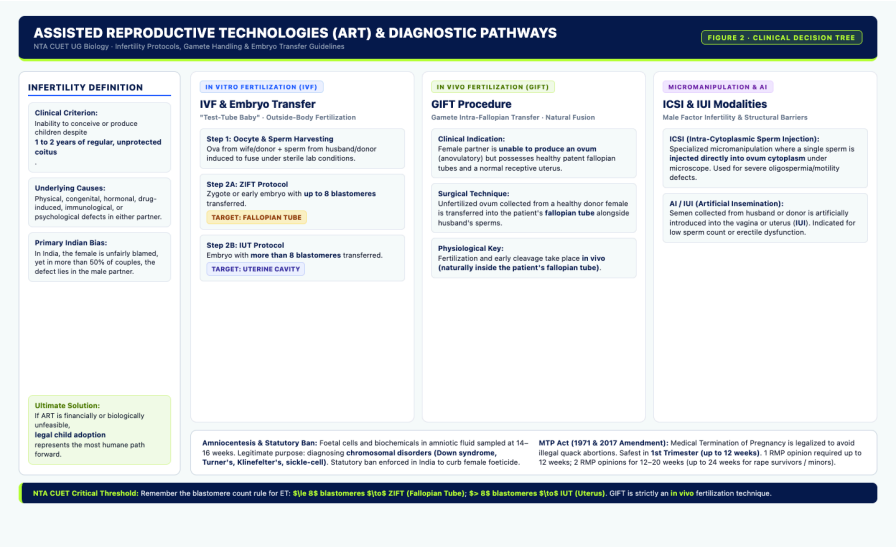


Figure 2: Clinical decision tree and technical workflow for Assisted Reproductive Technologies: categorizing in vitro fertilization (IVF: ZIFT vs IUT), in vivo gamete transfer (GIFT), and specialized micromanipulation (ICSI, IUI).

Specialized Protocols of Assisted Reproductive Technologies (ART)

- **1. In Vitro Fertilisation (IVF) & Embryo Transfer (ET) — "Test-Tube Baby":**
  - **Ovarian Stimulation & Harvesting:** The female partner undergoes controlled ovarian hyperstimulation using exogenous gonadotropins (FSH/LH analogs) followed by an hCG trigger. Multiple mature oocytes are harvested via ultrasound-guided transvaginal follicular aspiration.
  - **In Vitro Insemination:** Harvested ova and washed, capacitated spermatozoa are co-incubated in specialized culture dishes under simulated intrauterine conditions (37° C, 5%CO_2).
  - **Embryo Transfer (ET) Criteria (The Sacred Blastomere Boundary):**
    - **ZIFT (Zygote Intra-Fallopian Transfer):** The zygote or early cleavage-stage embryo containing **up to 8 blastomeres** is laparoscopically transferred directly into the **Fallopian Tube**. The embryo migrates down the tube naturally, entering the uterus as a blastocyst.
    - **IUT (Intra-Uterine Transfer):** Embryos cultivated beyond the 8-cell stage (such as 16-cell morula or early blastocyst with **more than 8 blastomeres**) are transferred directly into the **Uterine Cavity** via a transcervical catheter.
- **2. In Vivo Fertilisation & Embryo Lavage:**
  - Embryos conceived naturally inside the reproductive tract of a fertile donor female can be non-surgically flushed out (lavaged) and transferred into the uterus of an infertile recipient female who can nurture pregnancy.
- **3. GIFT (Gamete Intra-Fallopian Transfer):**
  - **Clinical Indication:** The female partner is **anovulatory** (fails to ovulate) but possesses patent, structurally normal fallopian tubes, healthy pelvic vasculature, and an intact hormonal receptive uterus.
  - **Procedure:** An unfertilized ovum retrieved from a healthy donor female is transferred into the patient's **fallopian tube** along with the husband's prepared spermatozoa.
  - **Critical Physiological Difference: GIFT is strictly an in vivo fertilization technique** — syngamy and early cleavage take place naturally inside the patient's own fallopian tube.
- **4. ICSI (Intra-Cytoplasmic Sperm Injection):**

- An advanced micromanipulation procedure wherein an embryologist immobilizes a single selected spermatozoon and **microinjects it directly into the cytoplasm of the harvested oocyte** using an ultra-fine glass micropipette under a high-magnification inverted microscope.
- **Indications:** Severe oligospermia, severe asthenozoospermia (immotile sperms), high teratozoospermia, or defective acrosomal reaction preventing zona pellucida penetration.

- **5. Artificial Insemination (AI) & Intra-Uterine Insemination (IUI):**
  - Indicated when the male partner cannot achieve vaginal insemination due to erectile dysfunction, or produces semen with low sperm count (**oligospermia**).
  - **Protocol:** Semen collected from the husband (**AIH — Artificial Insemination Husband**) or an anonymous screened donor (**AID — Artificial Insemination Donor**) is washed, centrifuged to concentrate motile sperms, and introduced via catheter into the vagina or directly into the uterine cavity (**IUI**).

Surrogacy & Legal Adoption Framework

- **Surrogacy & The Surrogacy (Regulation) Act, 2021:**
  - When a female partner possesses non-functional ovaries or a severely damaged, surgically absent, or deformed uterus, a **surrogate mother** carries the pregnancy to term on behalf of the genetic parents.
  - Under Indian law, **commercial surrogacy is strictly banned**. Only **altruistic surrogacy** (where no monetary compensation other than medical expenses and insurance is paid to the surrogate) is legally permissible for legally married, clinically certified infertile Indian couples.
- **Legal Child Adoption (CARA):**
  - While ART offers biological remedies, the procedures are emotionally exhausting, technologically intensive, and financially demanding, available only in specialized centers.
  - For millions of orphaned, destitute, or abandoned children in India, **legal adoption under the Central Adoption Resource Authority (CARA)** framework represents the most humane, legally established, and socially fulfilling solution for childless couples.

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High-Yield Formula & Concept Cheatsheet

Concept / Structure	Diagnostic Criterion	High-Yield Exam Trigger
Lactational Amenorrhea	High Prolactin → GnRH drop	Effective strictly up to a maximum of 6 months post-partum.
Copper IUD Triad	CuT, Cu7, Multiload 375	Suppresses sperm motility, viability, and fertilizing capacity.
Hormone IUD Duo	Progestasert, LNG-20	Renders uterus unsuitable for implantation and cervix hostile.
Saheli Pill	Centchroman (CDRI)	Non-steroidal, once-a-week oral pill with minimal side effects.
Emergency Contraception	Within 72 Hours	Progestogens or LNG-IUD administered post-unprotected coitus.
MTP First Trimester	Up to 12 Weeks	Safest abortion window; requires 1 RMP medical opinion.
Incurable STIs	Hep B, HSV-2, HIV	All other bacterial/protozoan STIs curable if treated early.
ZIFT vs IUT Boundary	8 Blastomeres Cutoff	≤ 8 cells → Fallopian Tube (ZIFT); > 8 cells → Uterus (IUT).
GIFT Modality	In Vivo Fertilization	Donor ovum transferred into recipient tube; fusion inside body.

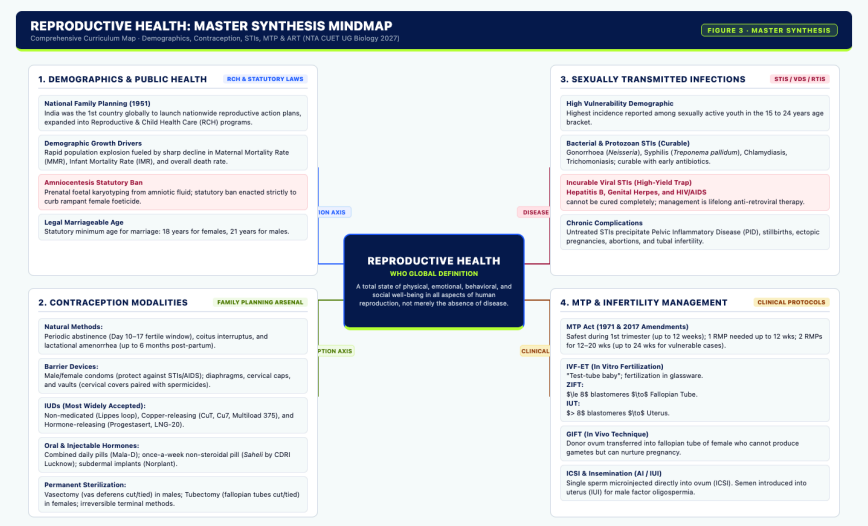


Figure 3: Master synthesis mindmap of Reproductive Health: integrating demographics, contraception spectrum, MTP acts, STI clinical pathology, and assisted reproductive technologies.

Practice Questions — Part 1 (Q1 to Q5)

Q1. Amniocentesis is a clinical procedure legally permitted in India solely for which of the following medical indications?

(A) Prenatal sex determination to maintain family gender balance
(B) Detecting foetal chromosomal aneuploidies such as Down syndrome and metabolic disorders
(C) Assessing maternal pelvic dimensions prior to vaginal delivery
(D) Inducing selective abortion of female foetuses in high-parity mothers

Sol. Q1. Amniocentesis is a prenatal diagnostic technique wherein amniotic fluid containing foetal skin and mucosal cells is aspirated to detect chromosomal abnormalities (e.g., Down syndrome, Turner syndrome, Klinefelter syndrome) and inborn metabolic disorders (e.g., PKU). Under the PCPNDT Act 1994, its use for prenatal sex determination is strictly banned to eradicate female foeticide. **Answer: (B)**

Q2. Which of the following represents an essential pharmacological attribute of an ideal contraceptive device?

(A) It must irreversibly block gametogenesis in both partners
(B) It must be user-friendly, highly effective, completely reversible, with minimal side effects
(C) It must permanently eliminate secondary sexual characteristics to suppress libido
(D) It must act strictly post-implantation by inducing chemical abortion

Sol. Q2. According to NCERT, an ideal contraceptive must be user-friendly, easily available, effective, completely reversible with no or least side effects, and must not interfere with the user's normal sexual drive, desire, or the sexual act. Contraceptives prevent conception, rather than inducing abortion post-implantation. **Answer: (B)**

Q3. Lactational amenorrhea functions as an effective natural birth control method primarily because:

(A) Placental progesterone persists in maternal circulation for up to 12 months
(B) Intense infant suckling elevates prolactin, suppressing pulsatile GnRH and gonadotropin secretion
(C) Maternal glycogen reserves are exhausted, inhibiting ovarian follicular angiogenesis
(D) Oxytocin release causes continuous uterine contractions that dislodge unfertilized ova

Sol. Q3. Post-partum intense suckling by the infant stimulates maternal prolactin secretion. Elevated prolactin exerts negative feedback on the hypothalamus, suppressing pulsatile GnRH release. Consequently, anterior pituitary FSH and LH surges are abolished, preventing follicular growth and ovulation. This natural method is effective for a maximum duration of 6 months post-partum. **Answer: (B)**

Q4. What is the precise cellular mechanism of action by which copper-releasing IUDs (such as CuT, Cu7, and Multiload 375) prevent pregnancy?

(A) They induce permanent surgical occlusion of the bilateral fallopian tubes
(B) Released copper ions suppress sperm motility and fertilizing capacity while stimulating phagocytosis of sperms
(C) They prevent ovulation by maintaining high circulating levels of progesterone
(D) They dissolve the zona pellucida of the secondary oocyte inside the ovary

Sol. Q4. Copper-releasing IUDs (CuT, Cu7, Multiload 375) continuously elute ionized copper (Cu^{2+}) into the uterine lumen. Copper ions suppress sperm motility, impair sperm flagellar kinematics, and destroy sperm fertilizing capacity, while the physical presence of the IUD promotes macrophage phagocytosis of spermatozoa in the uterine cavity. **Answer: (B)**

Q5. How do hormone-releasing intrauterine devices such as Progestasert and LNG-20 primarily execute their contraceptive function?

(A) By releasing cytotoxic heavy metals that destroy endometrial stroma
(B) By rendering the endometrium unsuitable for implantation and making the cervix hostile to sperms
(C) By chemically hydrolyzing the acrosome of mature spermatozoa inside the vagina
(D) By stimulating pituitary LH surge to accelerate premature luteinization

Sol. Q5. Hormone-releasing IUDs (Progestasert, LNG-20) slowly release progestin locally within the uterus. This induces endometrial glandular atrophy, rendering the endometrium completely unreceptive to blastocyst implantation, and thickens cervical mucus into a tenacious barrier hostile to sperm penetration. **Answer: (B)**

Practice Questions — Part 2 (Q6 to Q10)

Q6. The oral contraceptive pill 'Saheli', developed by scientists at CDRI Lucknow, is unique because it is:

(A) A steroidal pill containing high-dose testosterone taken daily for 21 days
(B) A non-steroidal formulation (Centchroman) taken once a week with minimal side effects
(C) An emergency pill that must be administered within 24 hours of coitus
(D) An injectable depot progestogen administered subcutaneously once every 5 years

Sol. Q6. Saheli (Centchroman / Ormeloxifene) is the world's first non-steroidal oral contraceptive pill, engineered by the Central Drug Research Institute (CDRI) in Lucknow, India. It acts as a selective estrogen receptor modulator that prevents blastocyst implantation. It is taken twice weekly for the first 3 months and once weekly thereafter, exhibiting negligible side effects. **Answer: (B)**

Q7. To prevent potential conception following unprotected intercourse, sexual assault, or contraceptive failure, emergency contraceptives must be administered within a strict window of:

(A) 24 hours of coitus
(B) 72 hours of coitus
(C) 7 days of coitus
(D) 14 days of coitus

Sol. Q7. Emergency contraceptives (such as progestogens alone, progestogen-estrogen combination pills, or insertion of an IUD) are clinically proven to be highly effective if administered within a strict threshold of **72 hours (3 days)** following unprotected intercourse, rape, or barrier malfunction. **Answer: (B)**

Q8. Following a successful bilateral vasectomy in a human male, which of the following physiological statements is completely ACCURATE?

(A) The patient immediately ceases to synthesize testosterone and loses secondary sex traits
(B) The volume of ejaculate is drastically reduced to zero
(C) Spermatogenesis continues in the testes, but the ejaculated semen lacks spermatozoa
(D) Spermatozoa are diverted into the ureters and excreted during micturition

Sol. Q8. Vasectomy involves cutting and ligating a small section of both vasa deferentia. Because Leydig cells remain completely intact, testicular testosterone secretion and male secondary sexual traits remain normal. Spermatogenesis continues in the seminiferous tubules, but the sperms cannot traverse the severed duct and are phagocytosed. Ejaculate volume remains almost normal, but contains zero sperms (azoospermia). **Answer: (C)**

Q9. Under the Medical Termination of Pregnancy (MTP) Amendment Act 2017 of India, up to how many weeks of gestation is an abortion considered safest and requires the opinion of only one Registered Medical Practitioner?

(A) Up to 6 weeks
(B) Up to 12 weeks (First Trimester)
(C) Up to 20 weeks
(D) Up to 28 weeks

Sol. Q9. Under the MTP Act, abortion is legally permitted and medically safest during the **first trimester (up to 12 weeks of pregnancy)**, requiring the diagnostic opinion of **one Registered Medical Practitioner**. Second-trimester terminations (12 to 20 weeks) carry significantly higher maternal morbidity and mandate the concurring opinion of **two Registered Medical Practitioners**. **Answer: (B)**

Q10. Which of the following groups contains EXCLUSIVELY sexually transmitted infections that are currently completely INCURABLE?

(A) Gonorrhoea, Syphilis, and Trichomoniasis
(B) Hepatitis B, Genital Herpes, and HIV/AIDS
(C) Chlamydiasis, Genital Warts, and Syphilis
(D) Trichomoniasis, Hepatitis B, and Gonorrhoea

Sol. Q10. According to NCERT Class 12 Biology, except for **Hepatitis B**, **Genital Herpes**, and **HIV infection**, all other sexually transmitted infections (such as Gonorrhoea, Syphilis, Chlamydiasis, and Trichomoniasis) are completely curable if diagnosed early and treated properly with appropriate antibiotics or antiparasitics. **Answer: (B)**

Practice Questions — Part 3 (Q11 to Q15)

Q11. Which specific pathogenic microorganisms are the causative agents of the venereal diseases Gonorrhoea and Syphilis, respectively?

(A) Treponema pallidum and Neisseria gonorrhoeae (B) Neisseria gonorrhoeae and Treponema pallidum
(C) Chlamydia trachomatis and Trichomonas vaginalis (D) Herpes Simplex Virus and Human Papillomavirus

Sol. Q11. Gonorrhoea is caused by the Gram-negative diplococcus bacterium **Neisseria gonorrhoeae**. Syphilis is caused by the spirochete bacterium **Treponema pallidum**. Both infections are bacterial, transmitted primarily via sexual intercourse, and respond to antibiotic therapies if initiated prior to tertiary complications. **Answer: (B)**

Q12. In clinical reproductive medicine, a married couple is formally diagnosed with infertility only when they are unable to conceive after a minimum cohabitation period of:

(A) 6 months of unprotected sexual intercourse (B) 1 to 2 years of regular, unprotected sexual intercourse
(C) 3 years of coitus accompanied by barrier contraceptives (D) 5 years of intermittent cohabitation

Sol. Q12. Infertility is defined clinically and epidemiologically as the failure to achieve clinical pregnancy after **1 to 2 years (minimum 12 to 24 months)** of regular, unprotected sexual intercourse without the use of any contraceptive measures. **Answer: (B)**

Q13. During an Assisted Reproductive Technology (ART) procedure involving In Vitro Fertilization (IVF), what determines whether ZIFT or IUT is performed?

(A) ZIFT is performed if sperm count is low; IUT is performed if ovum quality is poor (B) ZIFT transfers embryos up to 8 blastomeres into the fallopian tube; IUT transfers embryos with more than 8 blastomeres into the uterus
(C) ZIFT transfers embryos into the uterus; IUT transfers unfertilized gametes into the fallopian tube (D) ZIFT is an in vivo procedure; IUT is strictly an in vitro procedure

Sol. Q13. In Embryo Transfer (ET) following IVF, the developmental stage dictates the anatomical transfer locus: the zygote or early embryo with **up to 8 blastomeres** is transferred into the fallopian tube (**ZIFT — Zygote Intra-Fallopian Transfer**), whereas an embryo with **more than 8 blastomeres** is transferred directly into the uterine cavity (**IUT — Intra-Uterine Transfer**). **Answer: (B)**

Q14. Gamete Intra-Fallopian Transfer (GIFT) is the assisted reproductive technology specifically recommended for a female patient who:

(A) Has bilateral blocked fallopian tubes and an atrophic non-receptive uterus (B) Cannot produce functional ova, but possesses patent fallopian tubes and a suitable uterine environment for conception and development
(C) Produces normal ova but whose husband has complete azoospermia (D) Suffers from severe cervical incompetence preventing natural delivery

Sol. Q14. GIFT (Gamete Intra-Fallopian Transfer) is indicated for a female who is anovulatory (cannot produce ova) but possesses patent fallopian tubes and a healthy uterine environment capable of supporting gestation. An ovum from a healthy donor is transferred into her fallopian tube along with sperm, allowing fertilization and cleavage to occur **in vivo naturally within her body**. **Answer: (B)**

Q15. Intra-Cytoplasmic Sperm Injection (ICSI) is an advanced micromanipulation technique primarily employed to treat infertility caused by:

(A) Complete absence of fallopian tubes in the female partner (B) Severe male factor infertility including severe oligospermia, asthenozoospermia, or inability of sperm to penetrate the zona pellucida
(C) Failure of blastocyst hatching due to uterine fibroids (D) Immunological rejection of the embryo during the second trimester

Sol. Q15. ICSI (Intra-Cytoplasmic Sperm Injection) is a specialized in vitro micromanipulation procedure wherein a single spermatozoon is injected directly into the cytoplasm of an ovum using a microscopic micropipette. It is the premier clinical solution for severe male-factor infertility, such as extreme oligospermia, immotile sperms, or defective acrosomal enzyme release. **Answer: (B)**

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